
بِسْمِ اللَّهِ الرَّحْمَٰنِ الرَّحِيمِ

# DIGITAL IMAGE

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- **Digital image is the numerical record of the radiance leaving each of the pixels in each of the spectral bands**
- **The range of these values is represented in terms of a radiometric resolution that is normally 8 ( 0-255),10,11,… bits**

# WHY DIGITAL IMAGE?

- No digitization is required by user
- No info missing or change from provider to user's computer
- Data storage is easier
- No color loss during the storage
- Contains more info (> 3 bands)
- Flexible hard copy output (combinations)

# DIGITAL IMAGE DATA SCHEME

- **BAND INTERLEAVED BY PIXEL(BIP)**
- **BAND INTERLEAVED BY LINE (BIL)**
- **BAND SEQUENTIAL (BSQ)**
- **Run–length Encoding**

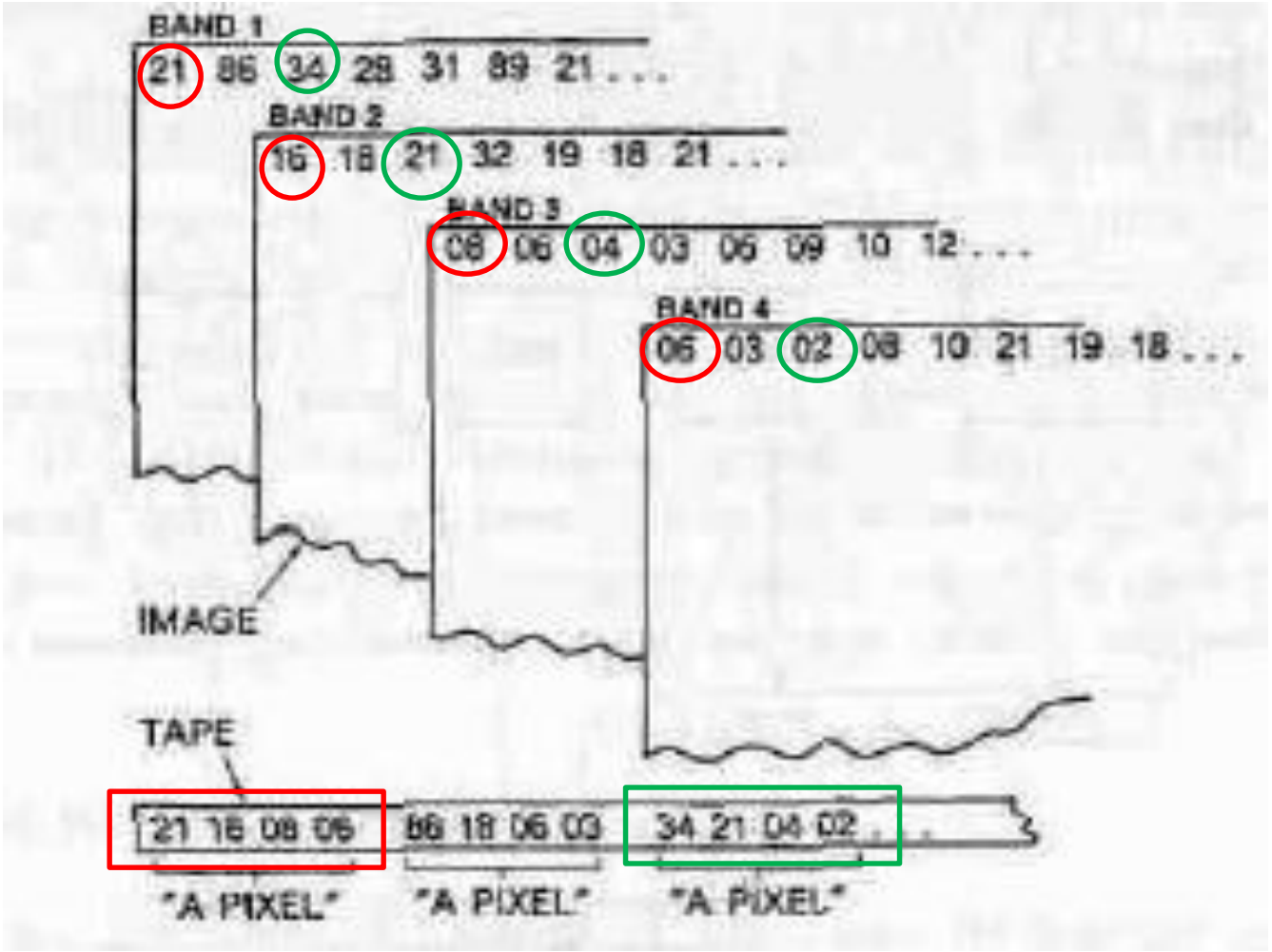
# Band Interleaved by Pixel (BIP)

- **Images stored in BIP format have**
  - **The first pixel for all bands in sequential order,**
  - **Followed by the second pixel for all bands,**
  - **Followed by the third pixel for all bands, etc., interleaved up to the number of pixels**

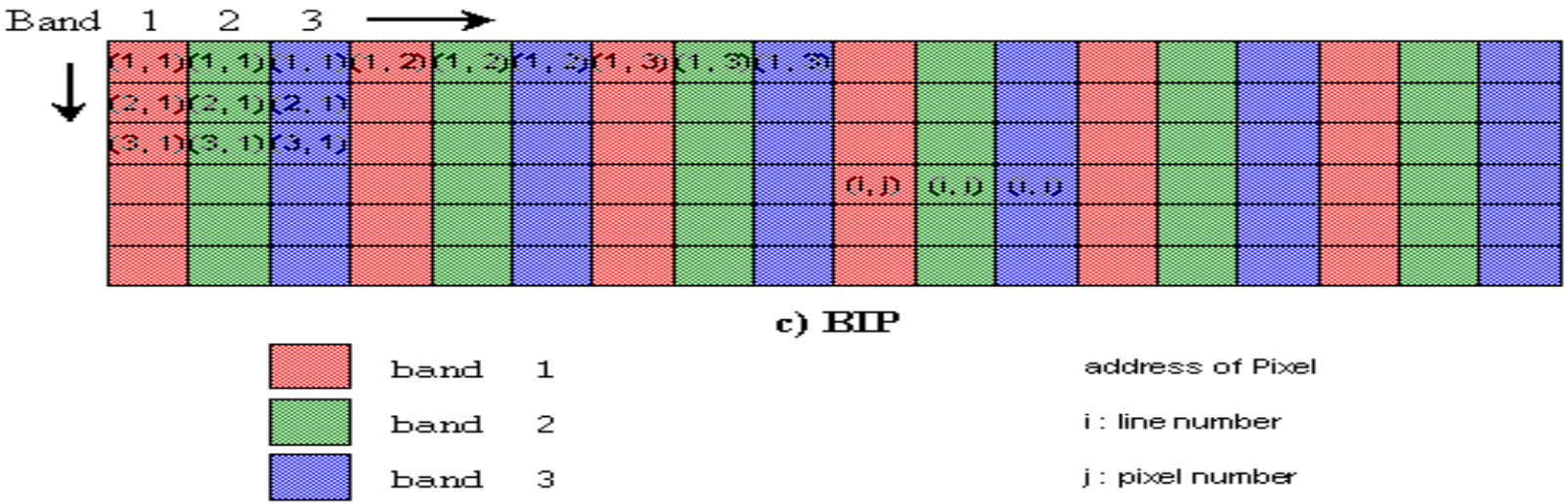
# BAND INTERLEAVED BY PIXEL (BIP)

- **In this format brightness values of each pixel are stored one after the other such as :-**
  - **Pixel (1,1) - band 1**
  - **Pixel (1,1) - band 2**
  - **Pixel (1,1) - band 3**
  - **Pixel (1,1) - band 4**
    - **Pixel (1,2) - band 1**
    - **Pixel (1,2) - band 2**
    - **Pixel (1,2) - band 3**
    - **Pixel (1,2) - band 4**

# BIP


BAND 1
21 86 34 28 31 89 21 . . .
BAND 2
16 18 21 32 19 18 21 . . .
BAND 3
08 06 04 03 06 09 10 12 . . .
BAND 4
06 03 02 08 10 21 19 18 . . .
IMAGE
TAPE
21 16 08 06
86 18 06 03
34 21 04 02
"A PIXEL"
"A PIXEL"
"A PIXEL"

# BIP Data Formats

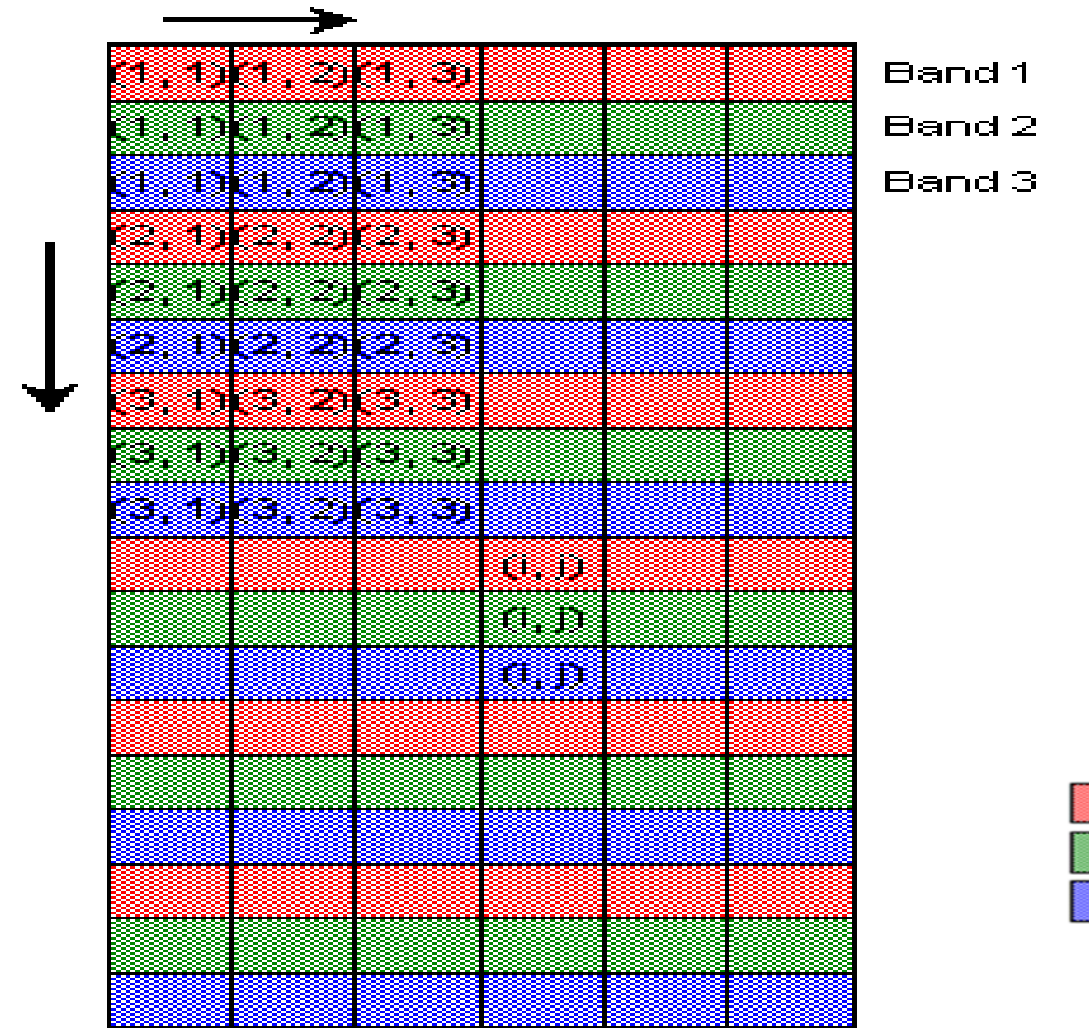



**For a given pixel, the values for all bands are side by side**

# BAND INTERLEAVED BY LINE (BIL)

- **It stores the first line of the first band followed by the first line of the second band, followed by the first line of the third band, interleaved up to the number of bands**
- **Subsequent lines for each band are interleaved in similar fashion**
- **Simply, data for the bands are written line by line**
  - **Line 1- band 1**
  - **Line 1- band 2**
  - **Line 1- band 3**
  - **Line 1 - band 4**
    - **Line 2- band 1**
    - **Line 2- band 2**
    - **Line 2- band 3**
    - **Line 2 - band 4**

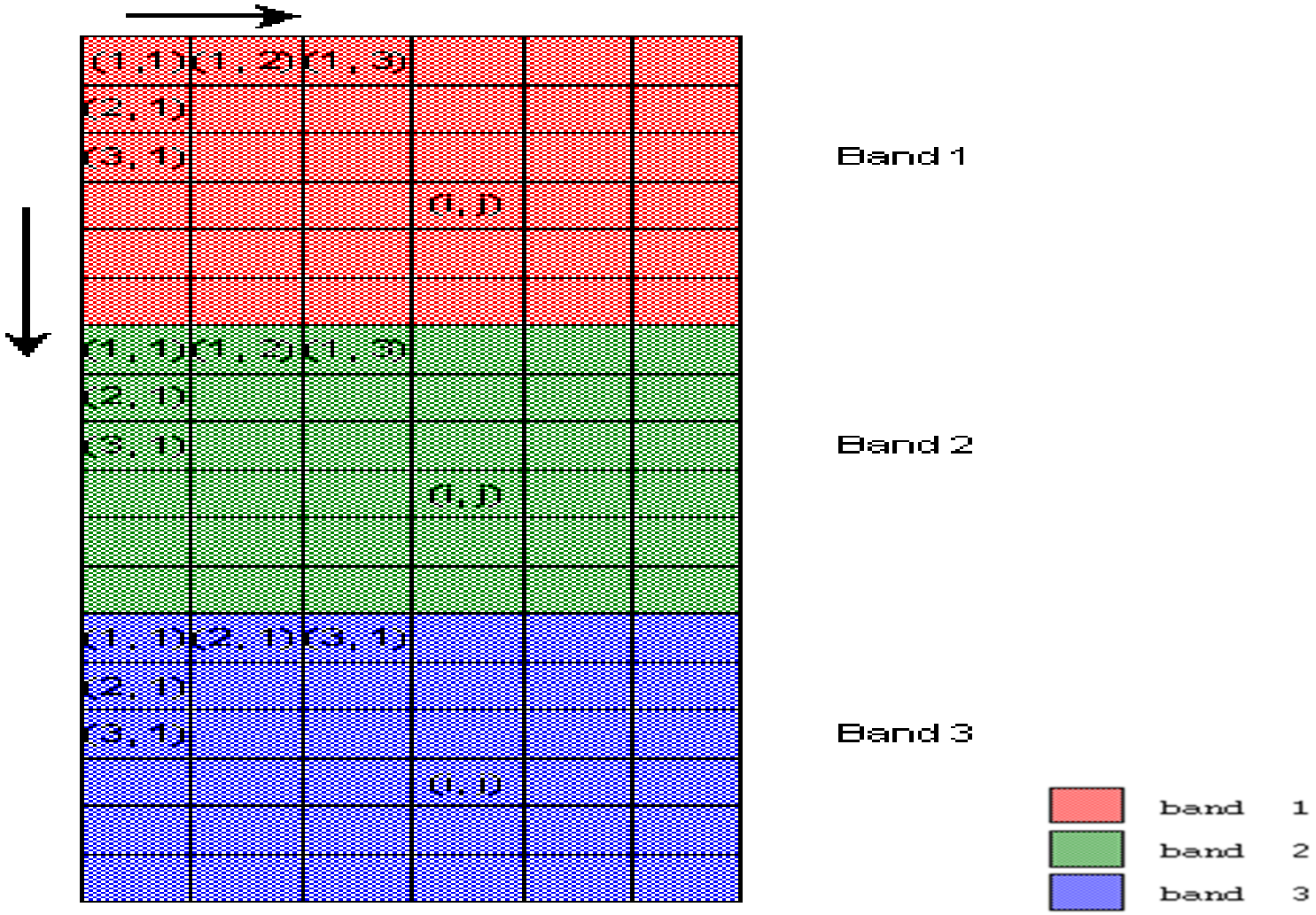
# BIL



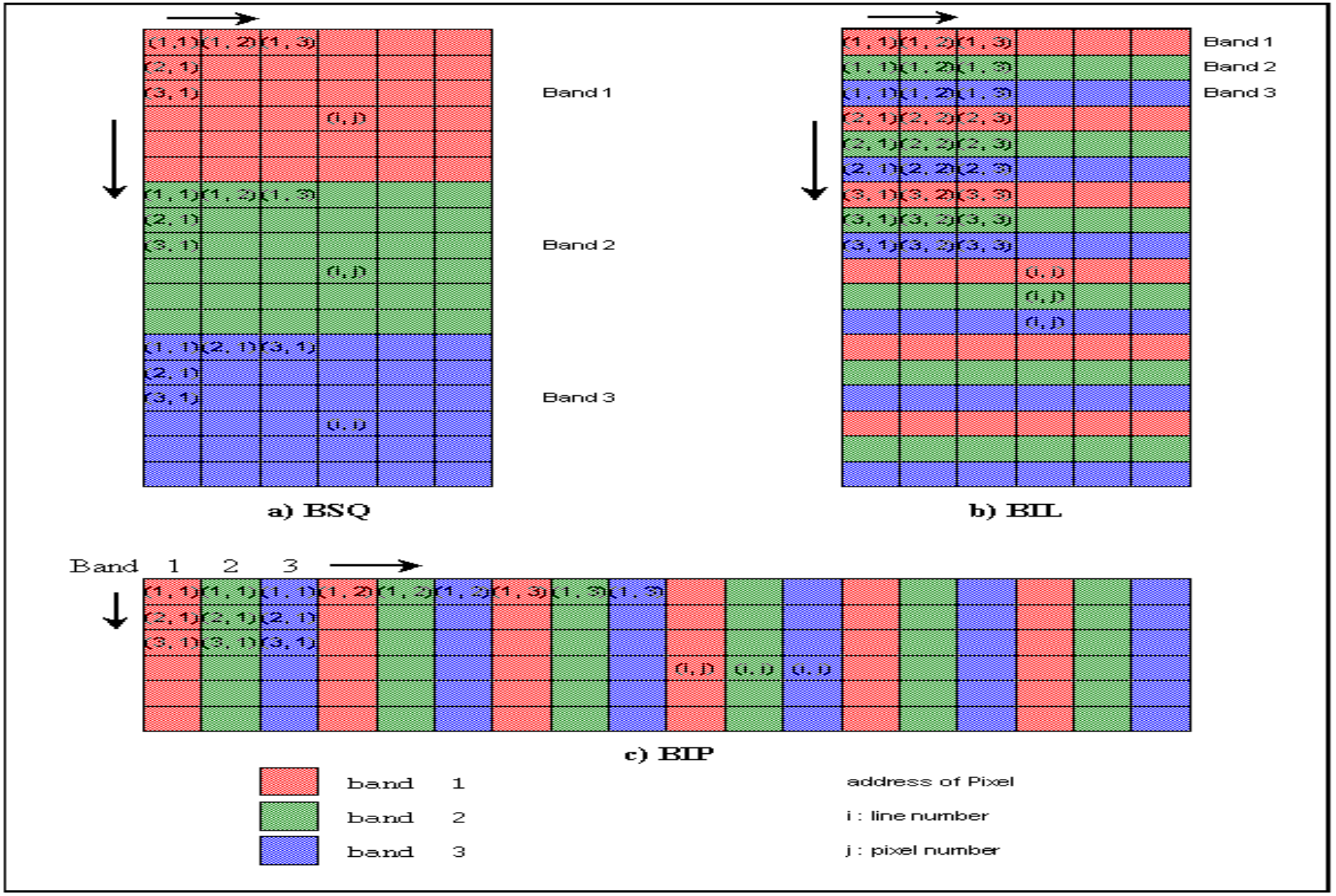
# BAND SEQUENTIAL (BSQ)

- All data for a single band covering entire scene is written as one file
- Each line of the data followed immediately by the next line in the same spectral band
- Each band of imagery is stored as a distinct file composed of a number of rows and columns

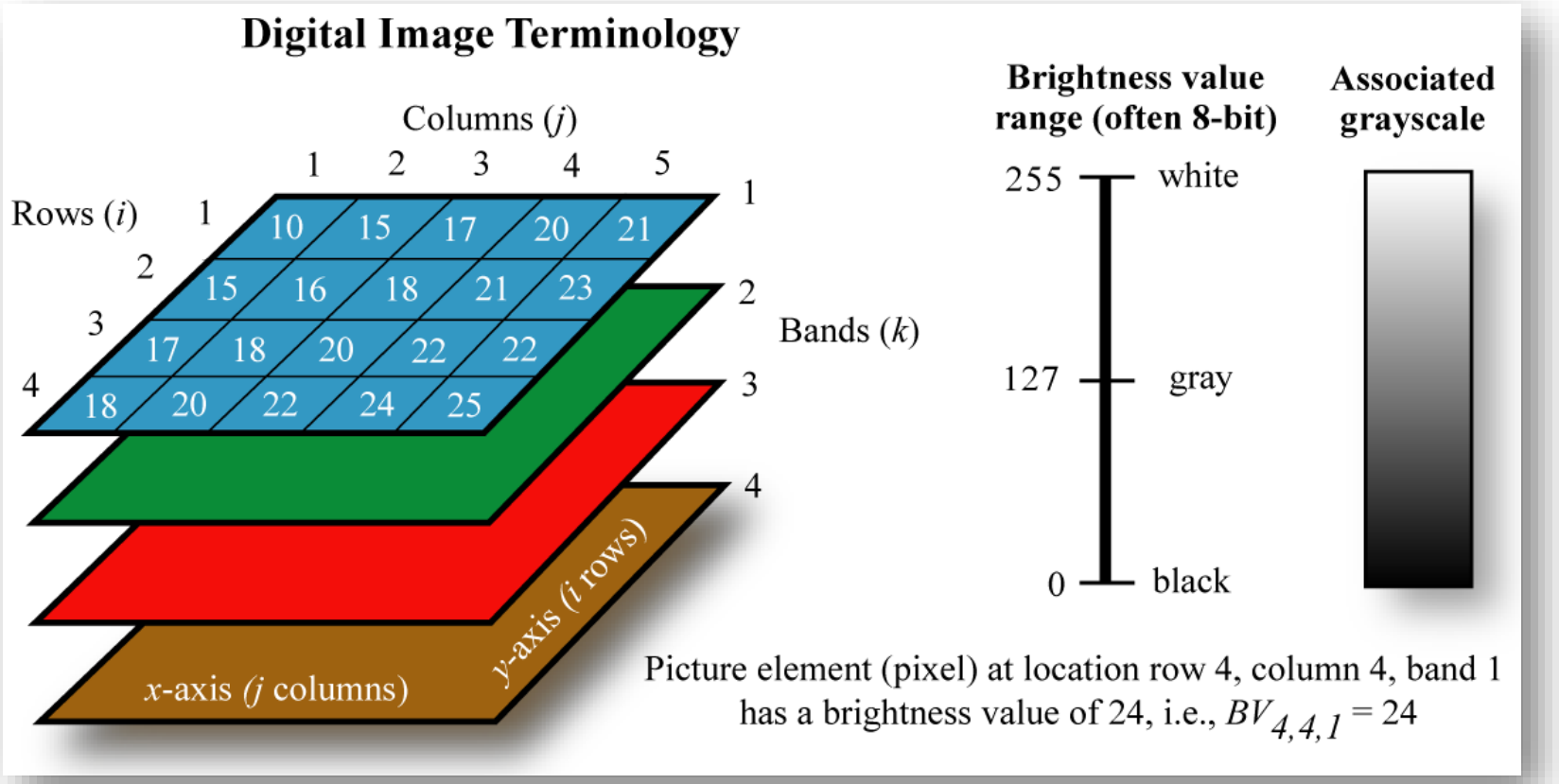
# BSQ


(1,1)(1,2)(1,3)
(2,1)
(3,1)
(i,j)
(1,1)(1,2)(1,3)
(2,1)
(3,1)
(i,j)
(1,1)(2,1)(3,1)
(2,1)
(3,1)
(i,j)
Band 1
Band 2
Band 3
band 1
band 2
band 3

# Data Formats



# Remote Sensing Raster (Matrix) Data Format




- **Each band is separate layer and can be visualized as four superimposed images**
- **Corresponding pixels in one band are registered exactly to those in other bands**

# Raster Data Storage

- **Values may be stored as:**
  - **Integer**
  - **Floating point**
- **In the case of floating points, pixels represent continuous data**
- **Floating point numbers typically use up more storage space**

# Image Data

- **Unsigned 8-bit** integers in the range [0, 255]
- **Unsigned 16-bit** integers in the range [0, 65535]
- **Signed 8-bit** integers in the range [-128, 127]
- **Signed 16-bit** integers in the range [-32768,  32767]
- **Floating Point:** support a much wider range of values

# Image Representations

- **Black and white image**
  - **Single color plane with 2 bits**
- **Grey scale image**
  - **Single color plane with more bits (8 bits)**
- **Color image**
  - **Three color planes each with 8 bits**

